

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1712

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

**Duration : 1 Hour
Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I Devadharshini M(192572149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:Devadharshini M

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Question 1: Apply Backtracking Search

Given Constraints

- D1 cannot work **Night**.
- D2 must work **before** D3 (Morning < Afternoon < Night).
- Only one doctor per shift.
- D3 cannot work **Morning**.
- Every doctor must be assigned exactly one shift.

Step 1: Initial Assignment

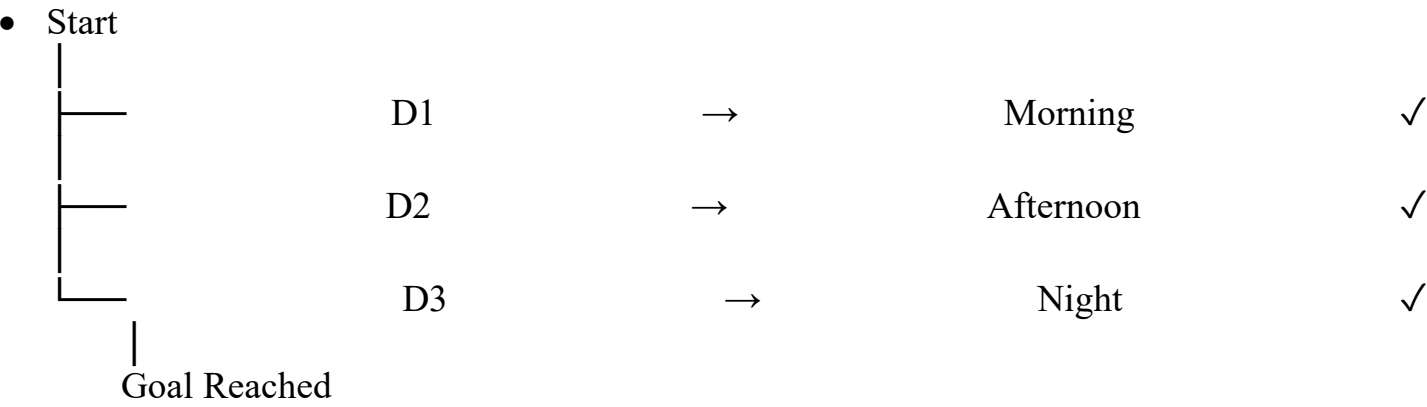
Doctor	Assigned Shift
D1	Morning
D2	Afternoon
D3	Night

Constraint Check

- D1 ≠ Night ✓
- D2 before D3 (Afternoon before Night) ✓
- One doctor per shift ✓
- D3 ≠ Morning ✓
- All doctors assigned ✓

- Since all constraints are satisfied, **Backtracking stops**.

Search Tree



Final Valid Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Conclusion: All constraints are satisfied. Therefore, the final schedule is:

- **Morning → D1, Afternoon → D2, Night → D3**

Question 2: BFS Search in a 5×5 Grid

BFS Algorithm

- Insert Start node (S) into the queue.
- Mark S as visited.
- Remove the front node from the queue.
- Expand its valid neighbours (Up, Down, Left, Right).
- Ignore blocked cells and visited nodes.
- Insert new nodes into the queue.
- Continue until Goal (G) is reached.

Manhattan Distance

- $h(n) = |x_1 - x_2| + |y_1 - y_2|$ $h(n) = |x_1 - x_2| + |y_1 - y_2|$

Example:

- Start = (1,1)
- Goal = (5,5)
- $h = |5 - 1| + |5 - 1| = 8$ $h = |5 - 1| + |5 - 1| = 8$ $h = |5 - 1| + |5 - 1| = 8$

Path Cost

- Each movement costs **1**.
- Total Cost = Number of moves.

Question 3: Rescue Robot Problem

(a) Constraint Analysis

- **Constraints**
 - Robot moves only Up, Down, Left and Right.
 - Cannot move through obstacles.
 - Can observe only adjacent cells.
 - Every move costs **1**.
 - Risky cells have an additional cost of **2**.
 - Robot must continuously update knowledge.
- **Effect on Decision Making**
 - Robot avoids blocked paths.
 - Chooses safer routes.
 - Updates its path whenever new obstacles are detected.
 - Prefers minimum-cost paths while ensuring safety.

(b) Solution Approach

- **Appropriate Search Strategy**
- **Uniform Cost Search (UCS) with Online Search Agent**
- **Steps**
 - Start at S.
 - Observe neighbouring cells.
 - Calculate movement cost.
 - Add risky-cell penalty if applicable.
 - Expand the node with the lowest cumulative cost.
 - Update knowledge after each move.
 - Repeat until Goal G is reached.

(c) AI Concepts

- **Intelligent Agent**
 - Senses the surrounding environment.
 - Makes decisions autonomously.
 - Performs actions to achieve the goal.
- **Rationality**
 - Selects the safest and least-cost path.
 - Avoids unnecessary movements.
- **Environment Type**
 - Partially Observable
 - Dynamic
 - Sequential
 - Discrete
 - Single-Agent

(d) Justification

- Uniform Cost Search combined with an Online Search Agent is the most suitable approach because:
 - It guarantees the least-cost path.
 - It handles different movement costs.
 - It updates decisions when new information becomes available.
 - It is ideal for disaster rescue operations where the environment changes continuously.
 - It ensures safe and efficient navigation.